

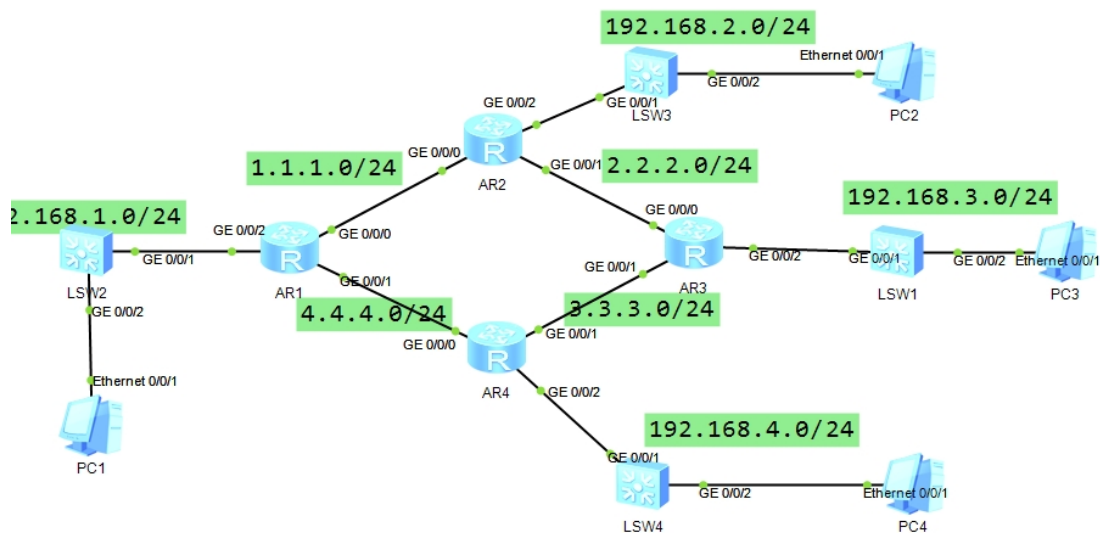
实验二：IP 路由技术

一、[实验目的]

- 1、掌握静态路由的配置；
- 2、掌握 RIP 和 OSPF 路由协议的配置。

二、[实验内容]

(一) 配置静态路由；(2 分)



R1:

```
[r1]ip route-static 192.168.2.0 24 1.1.1.21
[r1]ip route-static 192.168.3.0 24 1.1.1.21
[r1]ip route-static 192.168.4.0 24 4.4.4.41
[r1]ip route-static 2.2.2.0 24 1.1.1.21
[r1]ip route-static 3.3.3.0 24 4.4.4.41
```

R2:

```
[r2]ip route-static 192.168.1.0 24 1.1.1.12
[r2]ip route-static 192.168.4.0 24 1.1.1.12
[r2]ip route-static 192.168.3.0 24 2.2.2.32
[r2]ip route-static 3.3.3.0 24 2.2.2.32
[r2]ip route-static 4.4.4.0 24 1.1.1.12
```

R3:

```
[r3]ip route-static 192.168.1.0 24 2.2.2.23
[r3]ip route-static 192.168.2.0 24 2.2.2.23
[r3]ip route-static 192.168.4.0 24 3.3.3.43
[r3]ip route-static 1.1.1.0 24 2.2.2.23
```



```
[r3]ip route-static 4.4.4.0 24 3.3.3.43
R4:
[r4]ip route-static 192.168.1.0 24 4.4.4.14
[r4]ip route-static 192.168.2.0 24 4.4.4.14
[r4]ip route-static 192.168.3.0 24 3.3.3.34
[r4]ip route-static 1.1.1.0 24 4.4.4.14
[r4]ip route-static 2.2.2.0 24 3.3.3.34
```

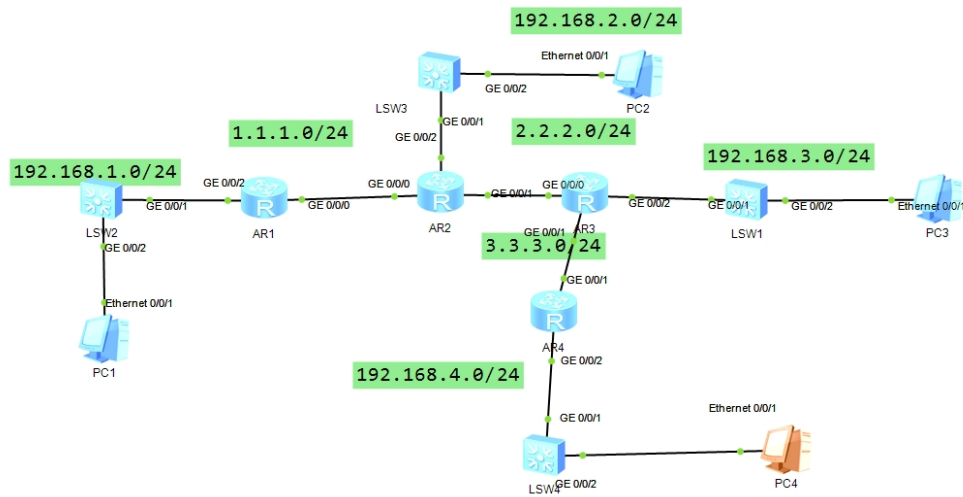
(二) 配置 RIP 路由协议。(1 分)

```
R1:
[r1]rip
[r1-rip-1]network 192.168.1.0
[r1-rip-1]net
[r1-rip-1]network 1.0.0.0
[r1-rip-1]net 4.0.0.0
R2:
[r2]rip
[r2-rip-1]net 192.168.2.0
[r2-rip-1]net 1.0.0.0
[r2-rip-1]net 2.0.0.0
R3:
[r3]rip
[r3-rip-1]net 192.168.3.0
[r3-rip-1]net 3.0.0.0
[r3-rip-1]net 2.0.0.0
R4: [r4]rip
[r4-rip-1]net 192.168.4.0
[r4-rip-1]net 4.0.0.0
[r4-rip-1]net 3.0.0.0
```

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(三) 配置 OSPF 路由协议。(2 分)





R1:

```
[r1]ospf 1
[r1-ospf-1]area 1
[r1-ospf-1-area-0.0.0.1]network 1.1.1.0 0.0.0.255
[r1-ospf-1-area-0.0.0.1]netw 192.168.1.0 0.0.0.255
```

R2:

```
[r2]ospf 1
[r2-ospf-1]area 0
[r2-ospf-1-area-0.0.0.0]network 192.168.2.0 0.0.0.255
[r2-ospf-1-area-0.0.0.0]area 1
[r2-ospf-1-area-0.0.0.1]network 1.1.1.0 0.0.0.255
[r2-ospf-1-area-0.0.0.1]area 2
[r2-ospf-1-area-0.0.0.2]network 2.2.2.0 0.0.0.255
```

R3:

```
[r3]ospf 1
[r3-ospf-1]area 2
[r3-ospf-1-area-0.0.0.2]netw 2.2.2.0 0.0.0.255
[r3-ospf-1-area-0.0.0.2]netw 3.3.3.0 0.0.0.255
[r3-ospf-1-area-0.0.0.2]netw 192.168.3.0 0.0.0.255
```

R4:

```
[r4]ospf 1
[r4-ospf-1]area 2
[r4-ospf-1-area-0.0.0.2]netw 3.3.3.0 0.0.0.255
```

